Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “system” to explain your model or result. Sentence starters are optional.

interaction: two objects affecting one another

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Bring back an earlier idea

With a fixed net force, a 4 kg system accelerates at 12 m/s².
Predict acceleration for a 8 kg system.
State a control needed for this comparison.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels 20 N from the other. If their masses are 40 and 80 kg, accelerations are 0.5 and 0.25 m/s² in opposite directions.

Your turn: Draw separate force diagrams for the horizontal interaction.
Explain why equal forces do not imply equal accelerations.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Name both forces

A hand pushes a wall to the right. Complete both statements: hand on____ pushes right;____ on hand pushes left.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

04 Heavy and light carts

A heavy cart and light cart push apart.
Compare the force sizes. Which has the greater acceleration from this interaction, and why?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Balanced or a pair?

A book rests on a table. Earth pulls book down; table pushes book up. Are these the two forces of one interaction pair?
Explain by naming the object receiving each.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

Start at 6 m, move to 16 m, then finish at 8 m.
Draw the route.
Find distance, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Afoot pushes ground backward.
Name the partner force and the object it acts on.
Explain how this can help the person move forward.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Choose two masses that would give a 3: 1 acceleration ratio while the forces remain equal.
Explain which skater accelerates more.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3